COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212	**SECTION: CPE31S2**	
DATE PERFORMED:	08/25/26	**DATE SUBMITTED:**	09/1/26
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ACTIVITY NO. 4.1 **Running Elevated Ad hoc Commands**

1. PROCEDURE OUTPUT

Task 1: Run elevated ad hoc commands

1.1

```
kaizohost@kaizohost-VirtualBox:~/ansible-lab$ ansible all -i inventory.yaml -m a
pt -a update_cache=true
server1 | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "msg": "Failed to lock apt for exclusive operation: Failed to lock directory
 /var/lib/apt/lists/: E:Could not open lock file /var/lib/apt/lists/lock - open
(13: Permission denied)"
}
server2 | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "msg": "Failed to lock apt for exclusive operation: Failed to lock directory
 /var/lib/apt/lists/: E:Could not open lock file /var/lib/apt/lists/lock - open
(13: Permission denied)"
}
```

```
kaizohost@kaizohost-VirtualBox:~/ansible-lab$ ansible all -i inventory.yaml -m a
pt -a update_cache=true -b -K
BECOME password:
server1 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "cache_update_time": 1788170340,
    "cache_updated": true,
    "changed": true
}
server2 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "cache_update_time": 1788170340,
    "cache_updated": true,
    "changed": true
}
```

1.2

```
kaizohost@kaizohost-VirtualBox:~/ansible-lab$ ansible all -i inventory.yaml -m apt -a name=vim-nox -b -K
BECOME password:
server2 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "cache_update_time": 1788170340,
    "cache_updated": false,
    "changed": true,
    "stderr": "",
    "stderr_lines": [],
    "stdout": "Reading package lists...\nBuilding dependency tree...\nReading state information...\nThe fo
es were automatically installed and are no longer required:\n  ansible-core libfwupd2 libgl1-amber-dri libg
linux-hwe-6.17-tools-6.17.0-40 linux-modules-6.17.0-40-generic\n  python3-argcomplete python3-dnspython py
 python3-kerberos\n  python3-libcloud python3-lockfile python3-ntlm-auth python3-packaging\n  python3-pass
quests-ntlm python3-resolvelib python3-selinux\n  python3-simplejson python3-winrm python3-xmltodict\nUse
remove' to remove them.\nThe following additional packages will be installed:\n  fonts-lato javascript-com
ry liblua5.1-0 libruby libruby3.2\n  rake ruby ruby-net-telnet ruby-rubygems ruby-sdbm ruby-webrick ruby-x
.2 rubygems-integration vim-common vim-runtime vim-tiny xxd\nSuggested packages:\n  apache2 | lighttpd | h
ev bundler cscope vim-doc indent\nThe following NEW packages will be installed:\n  fonts-lato javascript-c
uery liblua5.1-0 libruby libruby3.2\n  rake ruby ruby-net-telnet ruby-rubygems ruby-sdbm ruby-webrick ruby-
y3.2 rubygems-integration vim-nox vim-runtime\nThe following packages will be upgraded:\n  vim-common vim-
graded, 17 newly installed, 0 to remove and 47 not upgraded.\nNeed to get 18.4 MB/19.7 MB of archives.\nAf
tion, 83.8 MB of additional disk space will be used.\nGet:1 http://ph.archive.ubuntu.com/ubuntu noble/main
ato all 2.015-1 [2781 kB]\nGet:2 http://ph.archive.ubuntu.com/ubuntu noble/main amd64 javascript-common al
6 B]\nGet:3 http://ph.archive.ubuntu.com/ubuntu noble/main amd64 libjs-jquery all 3.6.1+dfsg+~3.5.14-1 [328
ttp://ph.archive.ubuntu.com/ubuntu noble/universe amd64 liblua5.1-0 amd64 5.1.5-9build2 [120 kB]\nGet:5 ht
e.ubuntu.com/ubuntu noble/main amd64 rubygems-integration all 1.18 [5336 B]\nGet:6 http://ph.archive.ubuntu
oble-updates/main amd64 ruby3.2 amd64 3.2.3-1ubuntu0.24.04.8 [50.7 kB]\nGet:7 http://ph.archive.ubuntu.com/
main amd64 ruby-rubygems all 3.4.20-1 [238 kB]\nGet:8 http://ph.archive.ubuntu.com/ubuntu noble/main amd64
```

```
kaizoserver1@kaizoserver1-VirtualBox:~$ which vim
/usr/bin/vim
```

```
kaizoserver2@kaizoserver2-VirtualBox:~$ which vim
/usr/bin/vim
```

a.

```
kaizoserver1@kaizoserver1-VirtualBox:~$ apt search vim-nox
Sorting... Done
Full Text Search... Done
vim-nox/noble-updates,noble-security,now 2:9.1.0016-1ubuntu7.20 amd64 [installed]
  Vi IMproved - enhanced vi editor - with scripting languages support

vim-tiny/noble-updates,noble-security,now 2:9.1.0016-1ubuntu7.20 amd64 [installed,automa
  Vi IMproved - enhanced vi editor - compact version
```

```
kaizoserver2@kaizoserver2-VirtualBox:~$ apt search vim-nox
Sorting... Done
Full Text Search... Done
vim-nox/noble-updates,noble-security,now 2:9.1.0016-1ubuntu7.20 amd64 [installed]
  Vi IMproved - enhanced vi editor - with scripting languages support

vim-tiny/noble-updates,noble-security,now 2:9.1.0016-1ubuntu7.20 amd64 [installed,automat
  Vi IMproved - enhanced vi editor - compact version
```

b. The logs show that server1 performed routine package management activities on the system, including updating existing system packages and installing new software. Some software was also removed shortly after being installed, while other installations included additional dependencies required for them to function.

1.3

```
kaizohost@kaizohost-VirtualBox:~/ansible-lab$ ansible all -i inventory.yaml -m apt -a name=snapd  -b -K
BECOME password:
server2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "cache_update_time": 1788170340,
    "cache_updated": false,
    "changed": false
}
server1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "cache_update_time": 1788170340,
    "cache_updated": false,
    "changed": false
}
```

```
kaizohost@kaizohost-VirtualBox:~/ansible-lab$ ansible all -i inventory.yaml -m apt -a "name=snapd state=latest"  -b -K
BECOME password:
server2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "cache_update_time": 1788170340,
    "cache_updated": false,
    "changed": false
}
server1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "cache_update_time": 1788170340,
    "cache_updated": false,
    "changed": false
}
```

Both commands were successful and showed changed: false because snapd is already installed and up-to-date on the remote servers. This means Ansible did not need to install or upgrade anything since the servers already match the desired state. The second command used double quotes to group the arguments together since it contains a space.

Task 2: Writing our First Playbook

2.1

install_apache.yml

```
1  ---
2  - hosts: all
3    become: yes
4    tasks:
5
6      - name: install apache2 package
7        apt:
8          name: apache2
9          state: present
```

2.2

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook -b -K install_apache.yml
BECOME password:

PLAY [all] *********************************************************************************************************

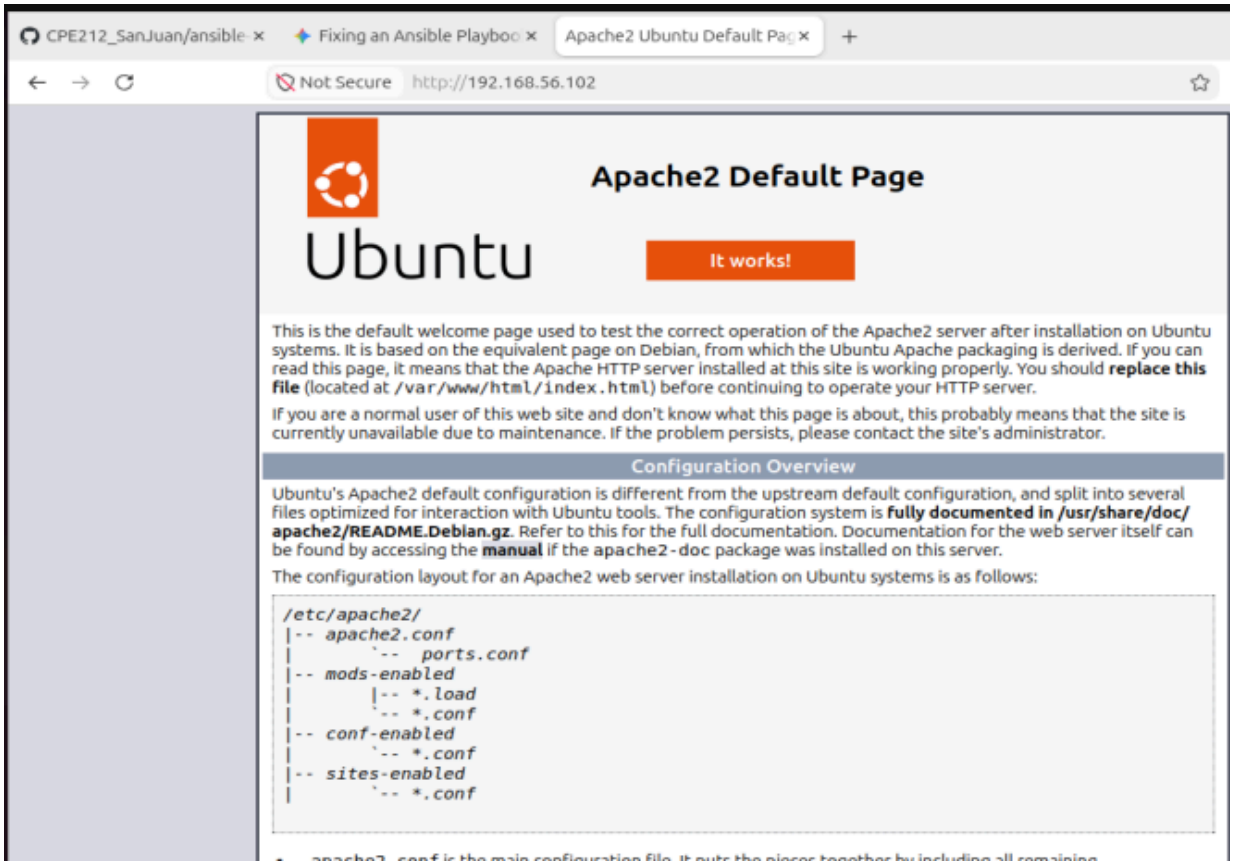
TASK [Gathering Facts] *********************************************************************************************
ok: [server2]
ok: [server1]

TASK [install apache2 package] *************************************************************************************
changed: [server2]
changed: [server1]

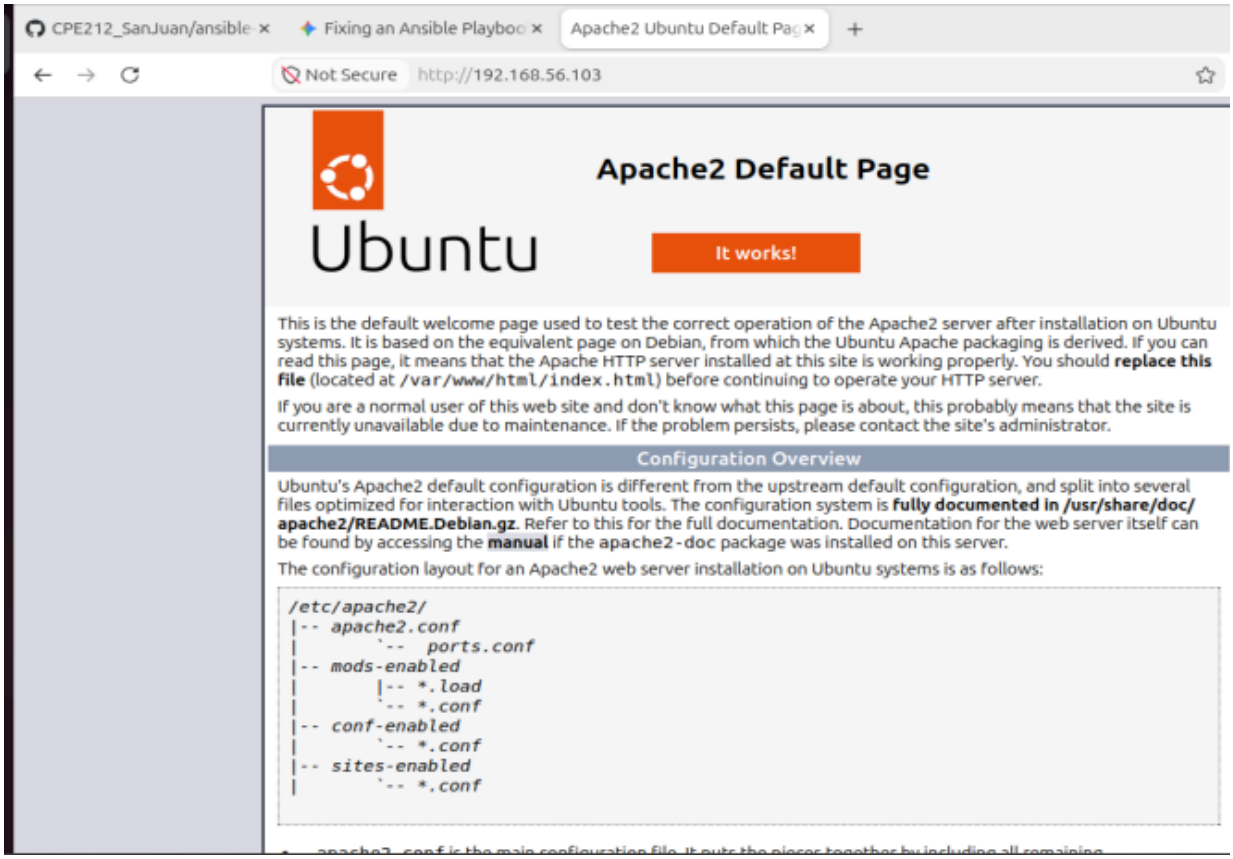
PLAY RECAP *********************************************************************************************************
server1                    : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

2.3

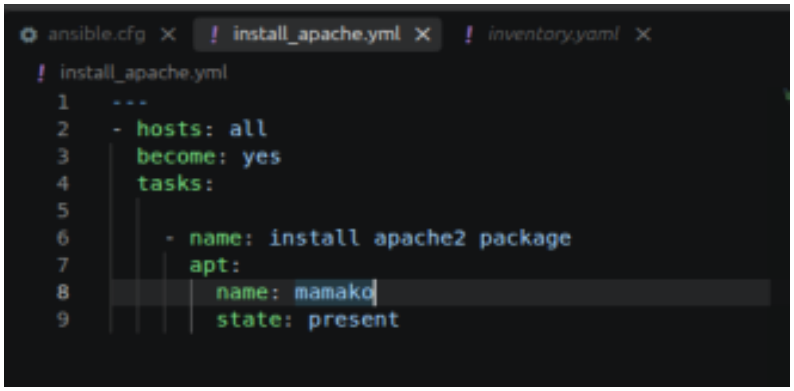
Server 1



Server 2


CPE212_SanJuan/ansible
Fixing an Ansible Playboo
Apache2 Ubuntu Default Pag
Not Secure
http://192.168.56.103
Apache2 Default Page
Ubuntu
It works!
This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.
If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.
Configuration Overview
Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the manual if the apache2-doc package was installed on this server.
The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:
/etc/apache2/
|-- apache2.conf
| `-- ports.conf
|-- mods-enabled
| |-- *.load
| `-- *.conf
|-- conf-enabled
| `-- *.conf
|-- sites-enabled
| `-- *.conf

2.4


ansible.cfg
install_apache.yml
inventory.yaml
install_apache.yml
1 ---
2 - hosts: all
3 become: yes
4 tasks:
5
6 - name: install apache2 package
7 apt:
8 name: mamako
9 state: present

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook -b -K install_apache.yml
BECOME password:

PLAY [all] ***********************************************************************************************************

TASK [Gathering Facts] ***********************************************************************************************
ok: [server1]
ok: [server2]

TASK [install apache2 package] ***************************************************************************************
fatal: [server2]: FAILED! => {"changed": false, "msg": "No package matching 'mamako' is available"}
fatal: [server1]: FAILED! => {"changed": false, "msg": "No package matching 'mamako' is available"}

PLAY RECAP ***********************************************************************************************************
server1                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
server2                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
```

2.5

install_apache.yml

```yaml
---
- hosts: all
  become: yes
  tasks:

    - name: update repository index
      apt:
        update_cache: yes

    - name: install apache2 package
      apt:
        name: apache2
        state: present

```

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook -b -K install_apache.yml
BECOME password:

PLAY [all] ***********************************************************************************************************

TASK [Gathering Facts] ***********************************************************************************************
ok: [server2]
ok: [server1]

TASK [update repository index] ***************************************************************************************
changed: [server2]
changed: [server1]

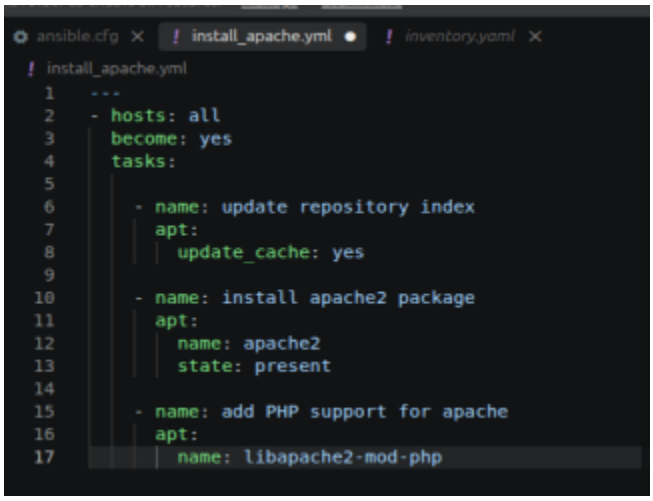
TASK [install apache2 package] ***************************************************************************************
ok: [server2]
ok: [server1]

PLAY RECAP ***********************************************************************************************************
server1                    : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored
server2                    : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored
```

2.6

Yes, the new update_cache command did change things on the remote servers. However, the install apache2 package command did not change anything because Apache was already installed from your previous run.

2.7




```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook -b -K install_apache.yml
BECOME password:

PLAY [all] ********************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************
ok: [server2]
ok: [server1]

TASK [update repository index] ************************************************************************************
changed: [server1]
changed: [server2]

TASK [install apache2 package] ************************************************************************************
ok: [server1]
ok: [server2]

TASK [add PHP support for apache] *********************************************************************************
changed: [server2]
changed: [server1]

PLAY RECAP ********************************************************************************************************
server1                    : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

2.8

Yes, the new command changed things. The add PHP support for apache task shows changed, meaning the libapache2-mod-php package was newly installed on both servers. The Update repository index task also shows changed. However, the install apache2 package task shows ok because it. was already installed from the previous run.

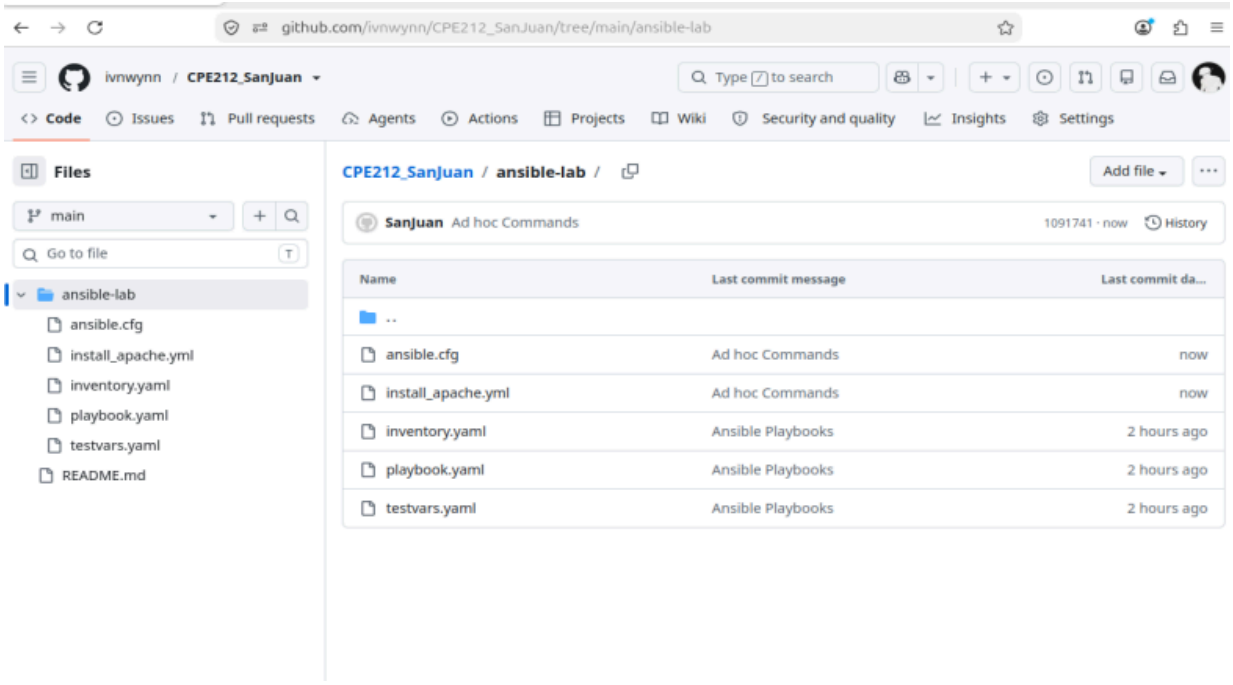
2.9

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   ansible.cfg

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        install_apache.yml

no changes added to commit (use "git add" and/or "git commit -a")
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ git add .
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ git commit -m "Ad hoc Commands"
[main 1091741] Ad hoc Commands
 2 files changed, 18 insertions(+), 1 deletion(-)
 create mode 100644 ansible-lab/install_apache.yml
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ git push
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 3 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 696 bytes | 696.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:ivnwynn/CPE212_SanJuan.git
   70496ab..1091741  main -> main
```



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the folowing:

1. What is the importance of using a playbook?
   - Playbooks automate repetitive tasks and ensure consistent configurations across multiple servers without manual error. They are also idempotent, meaning they can be run repeatedly without making unnecessary changes.
2. Summarize what we have done on this activity.
   - We created and executed an Ansible playbook that updates the repository cache, installs Apache2, and adds PHP support on both server1 and server2. We successfully verified the deployment by opening the firewall and confirming the Apache page loads in the browser.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **DEEPSEEK, GEMINI**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code assistance, and explanation of concepts**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I independently performed the original analysis, implementation, and final troubleshooting of the required work.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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